

Extra exercises: review

These exercises mix everything so far: SELECT, filtering and sorting, creating and changing tables, INSERT, UPDATE and DELETE, database design and JOINS. Before you start, check that `SELECT COUNT(*) FROM orders;` gives 15. If not, run `webshop_reset.sql` and press Ctrl+S.

Level 1

Use `webshop.db`.

Exercise 1

Show products in Clothing or Accessories that cost between 150 and 500 kr. Most expensive first.

Expected: 4 rows

Exercise 2

Which orders were placed in February 2026 and are not cancelled?

Expected: 4 rows

Exercise 3

Show every order line with the `order_id`, product name, quantity and line total (quantity times unit price). Only show lines where the line total is more than 500 kr. Biggest first.

Expected: 11 rows

Exercise 4

Which customers from Uppsala or Stockholm have placed at least one order? Each customer only once.

Expected: 5 rows

Level 2

These exercises change data. Click Revert Changes after each one, so `webshop.db` is back to normal.

Exercise 5

A new customer, Leo Falk from Uppsala, places an order today: 1 Hoodie Black and 2 Socks 3-pack. Add the customer, the order and the order lines. Then show the receipt with product names using a JOIN.

Expected: the receipt has 2 rows

Exercise 6

Order 12 is cancelled. Change its status, and put its products back in stock. Look at its order lines first to see which products and how many.

Expected: Sneakers Classic goes from 12 to 13 in stock, Socks 3-pack from 100 to 101

Exercise 7

Try to delete Hoodie Black (product 1). What happens, and why? How could the shop stop selling it without deleting it?

Expected: an error

Exercise 8

Add a column `discount_percent` to `products`. It should be 0 if nothing is given and can only be between 0 and 90. Give all Shoes 20 percent off. Then show every product with its price and its price after discount.

Look up: `ALTER TABLE ... ADD COLUMN` · **Expected:** 12 rows

Level 3

Exercises 9 to 12 use `webshop.db`. Exercise 13 is a small practice lab in a new database, `cinema.db`.

Exercise 9

Show every product with the dates it was ordered. Products that were never ordered must also be shown, with an empty date. Sort by product name.

Expected: 25 rows

Exercise 10

Which customers have bought something from the Shoes category? Each customer only once.

Expected: 3 rows

Exercise 11

Show all pairs of customers who live in the same city, for example Anna and Johan. Each pair should only appear once.

Look up: self join (a table joined with itself) · **Expected:** 5 pairs

Exercise 12

The price of Hoodie Black goes up to 649 kr. Change it, then find all order lines where the customer paid a different price than today's price. Show `order_id`, product name, what they paid and today's price.

Expected: 3 rows

Exercise 13

Practice lab. A cinema shows movies in two salons. Each movie has a title, a length in minutes and an age limit (0, 7, 11 or 15). A screening is one movie shown in one salon on a date and at a time. Customers buy tickets for screenings. Each ticket has a seat number and a price, and a seat can only be sold once per screening. Create `cinema.db`, draw the ER diagram and write `CREATE TABLE` with suitable constraints. Add test data: at least 3 movies, 4 screenings, 3 customers, 6 tickets, and one screening with no tickets. Then write three queries: all screenings with the movie title, sorted by date and time; every ticket with customer name, movie title and date; and the screenings that have no tickets sold.

Expected: depends on your own test data